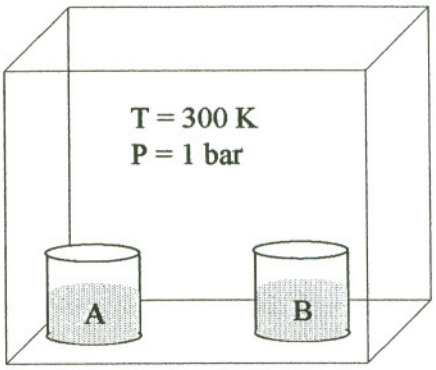
Physical Cume
November 15, 2002

Thermodynamics of Solutions and Phase Transition

Two bottles of aqueous solution (ideal) were open to the atmosphere inside a sealed container at 300 K and 1 bar. Solution A contains 0.1 mols of glucose in 2 mols of water, and solution B contains 0.2 mols of glucose in 2 mols of water. Assuming that glucose is completely non-volatile, the atmosphere in the container was saturated with water vapor before the bottles were open, and the total mass change of water in the vapor phase for the entire process was negligible.



(a) **(5+20 points))** Was the system in equilibrium when the bottles were just open in the container? Why?

(b) **(30 points)** If you think the system was not at equilibrium at the beginning, predict what will happen to the two solutions toward the equilibrium. What is the final equilibrium state going to be?

(c) **(25 points)** Calculate the amount of water in each solution at the final equilibrium state.

(d) **(5+15 points)** After the system reached the equilibrium predicted in (b) and (c), the temperature of the system underwent an isobaric increase to 320 K. Would the amount of water in each bottle be changed due to the temperature change? Why?

Hint: consider the chemical potential difference of the water in the two bottles.